Lithium acetate/PEG transformation of snowflake yeast

Adapted from the lab compendium Snowflake Yeast Protocols — Ratcliff Lab / MuLTEE (V3).

Reagents (per sample)

Reagent	Volume
PEG (50% w/w)	240 µL
LiOAc (1 M)	36 µL
ssDNA (2 mg/mL, freshly boiled)	50 µL
Digested plasmid/DNA	34 µL

Pre-culture

1. Inoculate 2 mL from an overnight culture into 10 mL fresh growth medium.
1. Grow the pre-culture for 3 hours.

Alternative: inoculate 1 mL from an overnight culture and grow for 4 hours instead of steps 1–2.

Cell prep

1. Transfer 1 mL of pre-culture into a labeled 1.5 mL tube (one tube per line/population). Vortex briefly before use.
2. Centrifuge at 8,000 rpm for 1 min.
3. Confirm pellet volume reaches approximately the 100 µL mark (minimum halfway to 100 µL). If insufficient, add another 1 mL of culture to the same tube and re-spin.
4. Remove supernatant without disturbing the pellet.
5. Add 100 µL water and pipette up and down to break up clusters until the mixture is evenly suspended (light cream color, no visible individual colonies). Later-generation clusters require more vigorous pipetting.
6. Add 900 µL water (1 mL total), centrifuge (8,000 rpm, 1 min), remove supernatant.
7. Wash with 1 mL water a second time (centrifuge, discard supernatant).
8. Incubate with 1 mL 0.1 M LiOAc (dilute stock, not the 1 M stock used in [[> Assembling the transformation mix]]) at 30°C for 1 hour.
9. Centrifuge and remove 900 µL of supernatant , pipetting to break up clusters.
10. Centrifuge and remove supernatant thoroughly.

ssDNA prep (concurrent with LiOAc washes)

1. Heat salmon sperm DNA (ssDNA) to 95 °C for 5 min.

11. Set heat block to 42 °C immediately after.
12. Mark the ssDNA tube each time it is boiled; do not reboil the same aliquot more than 3–4 times.
13. Recover boiled ssDNA on ice quickly before using at room temperature
14. Use ssDNA within ~30 min of boiling.

Assembling the transformation mix

Add to each cell pellet, in order:

1. **240 µL PEG (50% w/w)** — pipette slowly due to viscosity.

15. **36 µL 1 M LiOAc** — mix/swirl until uniformly cloudy.
    - Note specialized mixing technique:
      - Pipette up cells from bottom of solution and release at the top.
      - Swirl pipette around the sides of the tube on the way back down to the bottom to mix.
      - Repeat until large clusters crushed and solution uniform.
16. **50 µL freshly boiled ssDNA**.
17. **34 µL DNA solution** (plasmid, PCR product, etc.).
    - Swirl pipette around the sides of the tube on the way back down to the bottom to mix.
    - Invert 2-3 times

Heat shock and recovery

1. Incubate at **30 °C for 30 min** with rotation.

18. Heat-shock at **42 °C for 20 min** (heat block, ~350 rpm if available).
19. Centrifuge at 8,000 rpm for 2 min; remove all supernatant.
20. Resuspend in **1.0 mL YPD**, transfer to a fresh 2 mL tube, parafilm the lid closed to prevent lid popping open.
21. Recover at **30 °C for 4 hours**.
22. Plate the desired volume (typically 100–500 µL) on selective medium.